

Matthew Dula

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EDUCATION

Michigan State University

Ph.D. Electrical Engineering

East Lansing, MI

Expected 2028

Baylor University

B.S. Electrical and Computer Engineering - Summa Cum Laude

Waco, TX

May 2023

RESEARCH INTERESTS

RF signal processing and algorithms; distributed and integrated sensing and communication systems; algorithms for distributed computing and networks; software-defined radios (SDRs)

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug. 2023 – Present

Electromagnetics Research Group (EMRG)

Michigan State University, East Lansing, MI

- Major research topics include: high-accuracy wireless coordination of distributed phased array systems, algorithms for numerical optimization, hardware acceleration for wireless communication systems.
- Assisted development and maintenance for multiple Python/C++/GNU Radio libraries and software packages for software-defined radio, distributed array coordination, signal processing, and simulation.
- Mentored new group members on use and maintenance of DELTA group hardware and software.

Undergraduate Research Assistant

Aug. 2022 – Jan. 2023

Baylor Research and Innovation Collaborative (BRIC)

Baylor University, Waco, TX

- Investigated various statistical methods of detecting the presence of electromagnetic interference in protected frequency bands for weather radiometry.
- Designed and programmed feedback control loops to apply spectral-spatial masks to active devices to ensure regulatory compliance.

AWARDS AND FELLOWSHIPS

Honorable Mention, IEEE AP-S/URSI Student Paper Competition

June 2026

IEEE Antennas and Propagation Society

- For the paper "Localization of Distributed Phased Array Antennas in Partially Connected Networks Using Matrix Completion."

IEEE AP-S Graduate Fellowship

Jan. 2026

IEEE Antennas and Propagation Society

- For research towards "In-Situ Array Pattern Synthesis for Dynamic Distributed Arrays Using Wireless Localization."

Finalist, IEEE Wireless and Microwave Technology Conference Student Paper Competition

Apr. 2024

IEEE Florida West Coast Section

- For the paper "Decentralized Localization of Distributed Phased Array Elements Using High-Accuracy Ranging and Multidimensional Scaling."

Michigan State University ECE Distinguished Fellowship

Aug. 2023

Michigan State University, College of Engineering

- For "exceptional candidates in the top 5% of applicants who have identified a strong research topic match."

TEACHING AND MENTORING EXPERIENCE

Undergraduate Teaching Fellow

Jan. 2021 – May 2021

Introduction to Engineering Analysis (EGR 1302)

Baylor University, Waco, TX

- Held weekly office hours to provide supplemental lectures, field student questions, and review for exams.
- Collaborated with professors to grade weekly written homework assignments and MATLAB programs.
- Supported first-year students as they became accustomed to the university academic environment.

PROFESSIONAL EXPERIENCE

Manufacturing Equipment Engineering Intern

May 2021 – Aug. 2021

Qorvo, Inc.

Richardson, TX

- Designed and prototyped mechanical nitrogen release control systems for semiconductor spin-rinse dryers to reduce chemical waste during wafer cleaning, leading to 60% reduction in gas expenditure.
- Developed a shared inventory system for documenting maintenance and repairs of wafer production instruments, focusing on streamlining the connection from internal ownership to associated vendors.

PUBLICATIONS

Decentralized Localization of Distributed Antenna Array Elements Using an Evolutionary Algorithm

Matthew J. Dula, Naim Shandi, and Jeffrey A. Nanzer

Oct. 2025

IEEE Transactions on Microwave Theory and Techniques

doi: 10.1109/TMTT.2025.3564522

CONFERENCE PAPERS AND PRESENTATIONS

Localization of Distributed Phased Array Antennas in Partially Connected Networks Using Matrix Completion

Matthew J. Dula* and Jeffrey A. Nanzer

July 2026

2026 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting

doi:

- **Honorable Mention**, Student Paper Competition

Characterization of Near-Field Multi-Objective Beamforming in a Distributed Phased Array with Fully Wireless Coordination

Ahona Bhattacharyya*, Matthew J. Dula, and Jeffrey A. Nanzer

July 2026

2026 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting

Internode Ranging Using a Stepped-Frequency Coherent Pulse Train for Mitigation of Multipath Effects in Distributed Phased Arrays

Matthew J. Dula*, Anton Schlegel, and Jeffrey A. Nanzer

July 2025

2025 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting

doi: 10.23919/CNC-USNC-URSI64444.2025.11420133

Fully Wireless Collaborative Beamforming Using a Three-Element Coherent Distributed Phased Array

Jason M. Merlo*, Naim Shandi, Matthew J. Dula, Ahona Bhattacharyya, and Jeffrey A. Nanzer

Oct. 2024

IEEE International Symposium on Phased Array Systems and Technology (ARRAY)

doi: 10.1109/ARRAY58370.2024.10880359

- **2nd Place**, Student Paper Competition

Decentralized Localization of Distributed Phased Array Elements Using High-Accuracy Ranging and Multidimensional Scaling

Matthew J. Dula*, Naim Shandi, and Jeffrey A. Nanzer

Apr. 2024

IEEE Wireless and Microwave Technology Conference (WAMICON)

doi: 10.1109/WAMICON60123.2024.10522855

- **Finalist**, Student Paper Competition

UNIVERSITY SERVICE AND OUTREACH

MSU Prospective Students Day

October 2025

Presenter

- Prepared and staffed a booth and laboratory space to present our group's research work to an audience of prospective students, including a demonstration of a distributed synchronization experiment.

MSU Introduce a Girl to Engineering Program

June 2024, 2025

Volunteer

- Collaborated with the Michigan State University Women in Engineering Program to design and run events for high school students utilizing an X-band doppler radar and circuit design activities to spark engineering interest.

PROFESSIONAL SERVICE

Journal Reviewer:

- IEEE Journal of Lightwave Technology
- IEEE Internet of Things Journal

PROFESSIONAL MEMBERSHIPS

- Institute of Electrical and Electronics Engineers (IEEE), Graduate Student Member (2022 – Present)
- IEEE Young Professionals (2022 – Present)
- IEEE Aerospace and Electronic Systems Society (2026 – Present)
- IEEE Antennas and Propagation Society (2024 – Present)
- IEEE Microwave Theory and Technology Society (2024 – Present)

TECHNICAL SKILLS

Programming Tools and Skills:

- Languages:
 - Python (NumPy, Numba, SciPy, PyQt)
 - C/C++
- Software-Defined Radio:
 - GNU Radio, UHD, RFNoC
 - Ettus USRP X3x0
 - Some LiteX M2SDR and RFSoc 4x2
- Scientific Computing:
 - Vivado, Litex
 - Parallel computing
 - GPU computing
- Development:
 - Git, GitLab CI/CD
 - Linux
 - Pixi, Conda

Design Tools and Skills:

- Keysight ADS, Ansys HFSS, SPICE circuit simulation, microwave circuit design